Lecture quiz 1

Question 1 0.15 out of 0.15 points

What is an Operating System?

Selected Answer: All the other options are correct.

Answers:

Software program that controls other user programs.

Allocates and manages hardware resources.

Always ready to accept new commands from users and hardware.

All the other options are correct.

Question 2 0.15 out of 0.15 points

Why is there no transition from the waiting state to the running state?

Selected Answer: All the other options are correct.

Answers:

Because a process in the waiting state is unable to execute even if it is given access to the CPU.

Because once a process in the waiting state is ready to execute, it would first move to the ready state and then to the running state when scheduled fc execution by the OS.

Because this allows the OS to methodically select the process to execute from among all processes that are ready to execute on the CPU.

All the other options are correct.

Question 3 0.15 out of 0.15 points

What is the disadvantage of many-to-one mapping between logical (user) and physical (kernel) threads?

Selected Answer: A blocked user thread will block all the other user threads that are mapped to the same kernel thread.

Answers:

The mapping process is very complex.

A blocked user thread will block all the other user threads that are mapped to the same kernel thread.

There is no disadvantage.

A blocked user thread will also block all the other kernel threads in the system.

What comprises a process context and where is it stored?

Selected Answer: All registers required to execute/resume the process (program counter, stack pointer, memory protection registers, etc.) and it is stored in the PCB.

Answers:

All registers required to execute/resume the process (program counter, stack pointer, memory protection registers, etc.) and it is stored in the PCB.

The process memory as defined by the base and limit register values and it is stored in the user memory space.

All registers required to resume the process (program counter, stack pointer, memory protection registers, etc.) and it is stored in the user memory space.

Only base and limit register values for a process and it is stored in the PCB.

Question 5 0.15 out of 0.15 points

What is the difference between a long-term and medium-term scheduler?

Selected Answer: Long-term scheduler decides which new processes should be brought to main memory from disk; Medium-term scheduler decides which active processes to keep in main memory and which ones to swap to disk when load is heavy.

Answers:

Long-term scheduler schedules processes in main memory on the CPU; Medium-term scheduler decides which active processes to keep in main memo and which ones to swap to disk when load is heavy.

They are identical.

Long-term scheduler decides which new processes should be brought to main memory from disk; Medium-term scheduler schedules processes in mair memory on the CPU.

Long-term scheduler decides which new processes should be brought to main memory from disk; Medium-term scheduler decides which active processes to keep in main memory and which ones to swap to disk when load is heavy.

Question 6 0.15 out of 0.15 points

What parts of the process memory space are shared between threads in the same process.

Selected Answer: Text, Data, Heap and OS resources (files, etc.) are shared between the threads; Stack and Registers are not.

Answers:

Everything is shared between threads in a single process.

Only Text and OS resources (files, etc.) are shared; all other parts are private to the threads.

Text, Data, Heap and OS resources (files, etc.) are shared between the threads; Stack and Registers are not.

Text, Data and Heap are shared between the threads; OS resources (files, etc.), Stack and Registers are not.

Question 7

0.15 out of 0.15 points

How do you distinguish between direct and indirect message passing for inter-process communication?

Selected Answer: In direct message passing, communication occurs between specific processes; In indirect message passing, any process with access to a mailbox can send/receive messages to/from that mailbox.

Answers:

In direct message passing, communication occurs between specific processes; In indirect message passing, any process with access to a mailbox can send/receive messages to/from that mailbox.

In direct message passing, communication occurs between specific processes through shared user space memory; In indirect message passing, communication occurs through mailboxes.

Direct message passing is the same shared memory; Indirect message passing uses the kernel memory space.

Question 8

0.15 out of 0.15 points

What is an advantage and a disadvantage of many-to-many mappings of logical (user) to physical (kernel) threads?

Selected Answer: High concurrency and not many kernel threads required; mapping is very complex.

Answers:

High concurrency and mapping is very simple many kernel threads would be required.

High concurrency and not many kernel threads required; mapping is very complex.

Mapping is very simple and not many kernel threads required; concurrency is low.

High concurrency; mapping is very complex and many kernel threads would be required.

Question 9

0 out of 0.15 points

Which of the following precisely describes the sequence of steps to process an interrupt.

Selected Answer: Disable interrupts, save program context, locate interrupt service routine (ISR) in interrupt vector table, execute ISR, restore program context, enable interrupts.

Answers:

Disable interrupts, locate interrupt service routine (ISR) in interrupt vector table, execute ISR, enable interrupts.

Switch to kernel mode, save program context, locate interrupt service routine (ISR) in interrupt vector table, execute ISR, restore program context, switch to user mode.

Switch to kernel mode, locate interrupt service routine (ISR) in interrupt vector table, execute ISR, switch to user mode.

Disable interrupts, save program context, locate interrupt service routine (ISR) in interrupt vector table, execute ISR, restore program context, enable interrupts.

Question 10

Kernel or monitor or supervisor mode is the same as "superuser" in Linux.

Selected Answer: False: Kernel mode is a hardware mode of operation, whereas "superuser" is the "root" user account.

Answers:

False: Kernel mode is a hardware mode of operation, whereas "superuser" is the "root" user account.

True: Both, the kernel mode as well as "superuser" allows one to execute privileged instructions in hardware.

False: "superuser" is more privileged than kernel mode.

False: "superuser" is the same as supervisor mode, whereas kernel or monitor mode is more privileged.

Question 8

0.15 out of 0.15 points

There is one base and one limit register in the hardware for each user program.

Selected Answer: False: There is only one base and one limit register overall in the hardware, which are loaded with appropriate values by the OS before executing the user program.

Answers:

True: This is necessary to ensure memory protection for each user program.

False: There is only one base and one limit register overall in the hardware, which are loaded with appropriate values by the OS before executing the user program.

False:There is only one base and one limit register overall in the hardware, which are loaded with appropriate values by the CPU before executing the user program.

False: These registers are in the OS, which also checks whether each memory access is within range.

Question 9

0.15 out of 0.15 points

Multiprogrammed systems allow parallel execution of jobs, hence they are also called time-shared systems.

Selected Answer: False: Parallel execution of jobs depends on the hardware (multiprocessing); multiprogramming allows efficient switching between jobs for CPU access (time-division access), and hence it is also called time-shared system.

Answers:

False: Parallel execution of jobs depends on the hardware (multiprocessing); multiprogramming allows efficient switching between jobs for CPU access (time-division access), and hence it is also called time-shared system.

True: Multiprogramming allows simultaneous access to the CPU for multiple user jobs, and hence it is also called time-shared system.

Question 1 0.15 out of 0.15 points

What are the two typical execution orders in fork() system call

Selected Answer:	Parent process can execute in parallel to the forked child processes; Parent process can wait for the forked child processes to terminate before proceeding
Answers:	Forked child processes wait for parent process to terminate; Parent process can wait for the forked child processes to terminate before proceeding.
	Parent process can execute in parallel to the forked child processes; Parent process can wait for the forked child processes to terminate before proceeding
	Parent process executes before forked child processes; Forked child processes execute before parent process can continue.

Question 2 0.15 out of 0.15 points

What is the difference between a process and a program?

Selected Answer:	Process is an instance of a program that is currently active in the system
Answers:	They are identical
	Every user program can have at most one active process in the system.
	They are two different and unrelated concepts.
	Process is an instance of a program that is currently active in the system

Question 3 0.15 out of 0.15 points

Describe one advantage and one disadvantage of batch systems.

Selected Answer:	Memory management is simple (1 user job); CPU utilization is not efficient (idling during I/O).
Answers:	Memory management is simple (1 user job); CPU utilization is not efficient (scheduling of jobs is complex).
	Memory management is simple (1 user job); CPU utilization is not efficient (idling during I/O).
	CPU utilization is very efficient (1 user job to schedule); Memory management is complex (deciding which user job to keep in memory).

Question 5 0.15 out of 0.15 points

What is the difference between shared memory and message passing based Inter-Process Communication.

Selected Answer:	In message passing, the communication is handled by the OS; In shared memory, the communication is handled directly by the processes through a comm memory space.
Answers:	They are two different names for the same communication mechanism which uses kernel memory space.
	In message passing, the communication is handled by the OS; In shared memory, the communication is handled directly by the processes through a comm memory space.
	In shared memory, there is memory in the kernel space that is shared between the processes; In message passing, no memory is required in the kernel memory space.
	They are two different names for the same communication mechanism which uses user memory space.

Question 6 0.15 out of 0.15 points

What is an advantage and a disadvantage of one-to-one mapping of logical (user) and physical (kernel) threads?

Selected Answer:	More concurrency than many-to-one mappings; requires creation of several kernel threads (one per user thread).
Answers:	More concurrency than many-to-one mappings; mapping is very complex.
	More concurrency than many-to-many mappings; requires creation of several kernel threads (one per user thread).
	More concurrency than many-to-one mappings; requires creation of several kernel threads (one per user thread).
	Mapping is very simple; less concurrency than many-to-one mappings.

Question 7 0.15 out of 0.15 points

What are the typical resource constraints of embedded systems?

Selected Answer:	All the other options are correct.
Answers:	Small memory and low power (battery operated).
	Low network bandwidth and small displays.
	Slow processors
	All the other options are correct.

Question 10

0.15 out of 0.15 points

The following sequence of steps precisely defines the processing of a system call

Selected Answer: CPU generates a trap, CPU switches to kernel mode and transfers control to the OS, OS identifies the appropriate function for the system call and executes it, upon function completion OS switches the mode back to user mode and transfers control to the user program.

Answers:

OS generates a trap, OS switches to kernel mode, OS identifies the appropriate function for the system call and executes it, upon function completion OS switches the mode back to user mode and transfers control to the user program.

CPU generates a trap, CPU switches to kernel mode, CPU identifies the appropriate function for the system call and executes it, upon function completion CPU switches the mode back to user mode and transfers control to the user program.

CPU generates a trap, CPU switches to kernel mode and transfers control to the OS, OS identifies the appropriate function for the system call and executes it, upon function completion OS switches the mode back to user mode and transfers control to the user program.

CPU generates a trap, CPU switches to kernel mode and transfers control to the OS, OS identifies the appropriate function for the system call and executes it, upon function completion CPU switches the mode back to user mode and transfers control to the user program.

Question 1

0.15 out of 0.15 points

What are the four memory regions of a process and how are they used.

Selected Answer: "Text" for instructions; "Data" for static and global variables; "Stack" for managing function calls and local function variables; "Heap" for dynamically created variables.

Answers:

"Text" for documentation about the process; "Data" for all variables; "Stack" for managing function calls; "Heap" for garbage collection.

"Code" for instructions; "Global" for static and global variables; "Function" for managing function calls and local function variables; "Dynamic" for dynamical created variables.

"Text" for instructions; "Data" for static and global variables; "Stack" for managing function calls and local function variables; "Heap" for dynamically created variables.

All the other options are correct.

Question 5

In real-time systems, average response time of user programs should be minimized.

Selected Answer: True: This will ensure that the system is responsive.

Answers:

True: This will ensure that the required deadlines are met.

True: This will ensure that the system is responsive.

False: Real-time systems require guaranteed satisfaction of deadlines, irrespective of average response time.

False: Real-time systems require average response times to be maximized.

Question 8

DMA does not use interrupts.

Selected Answer: False: DMA uses interrupts once the block transfer is complete (between memory and I/O device) to notify the CPU.

Answers:

False: DMA uses interrupts once the block transfer is complete (between memory and I/O device) to notify the CPU.

True: DMA bypasses the CPU and hence does not require interrupts.

Question 10

System calls are identical to library function calls in a high-level programming language such as C.

Selected Answer: False: User programs can access services in the OS by invoking software functions implemented in the OS through system calls.

Answers:

True: For example, fopen() in C programming is a system call to open a file.

False: User programs can access services in the OS by invoking software functions implemented in the OS through system calls.